AI-DCGs Dataset Description

AI-DCGs is a multimodal AI-ready dataset for glacier semantic segmentation in High Mountain Asia. It provides Sentinel-2 optical reflectance, Sentinel-1 synthetic aperture radar, Landsat land surface temperature, Copernicus DEM terrain variables and pixel-level reference labels. The data are released at three patch sizes: 256×256, 512×512 and 1024×1024 pixels. Each sample comprises four feature GeoTIFFs and one label GeoTIFF, with 15 predictor channels in total. The dataset supports bare-ice and debris-covered-ice identification, multimodal fusion, feature ablation and comparisons of patch size.

1 Data coverage

The data originate from 42 sampling windows distributed across 13 glacier subdivisions in High Mountain Asia, including the Tien Shan, Pamir, Hindu Kush, Karakoram, Kunlun Shan, Tibetan interior, Nyainqentanglha, Himalaya and Hengduan Shan. The combined source-window extent is approximately 71.34–101.99° E and 27.58–43.68° N. The released patches cover only the retained portions of these windows after cleaning and do not form a continuous image mosaic of this bounding region. Reference-label years span 2016–2025. Remote-sensing features are processed or composited from multiple scenes for each window; the DEM is a static terrain variable.

2 Data releases and sample counts

Archive	Size in pixels	Training	Test	Total	Size in GiB
patches_256.zip	256×256	19,979	471	20,450	38.89
patches_512.zip	512×512	6,088	135	6,223	45.72
patches_1024.zip	1024×1024	1,841	36	1,877	162.15

One sample comprises 5 .tif files with the same split and numeric ID. Together, the three releases contain 28,550 patch samples and 142,750 GeoTIFF files. Different patch sizes cover shared source areas, so these samples cannot be treated as independent, non-overlapping observations.

3 File organization and labels

Each ZIP contains train and test directories at its root. Extract the contents into a directory named for the patch size. In the layout below, N is 256, 512 or 1024, and id is the patch identifier.

```
patches_<N>_clean/
  train/
    s2/<id>.tif
    s1/<id>.tif
    lst/<id>.tif
    dem/<id>.tif
    label/<id>.tif
  test/
    s2/<id>.tif
    s1/<id>.tif
    lst/<id>.tif
    dem/<id>.tif
    label/<id>.tif
```

Within each split, s2, s1, lst, dem and label files are paired by identical filenames. For example, train/s2/1.tif and train/label/1.tif refer to the same patch. Training and test IDs each start at 1.tif. Identical IDs in different splits or patch-size releases do not identify the same spatial sample.

3.1 Label codes

Pixel value	Class	Meaning
0	Background	Pixels not labelled as glacier in the reference mask
1	Bare ice	Glacier surface assigned to the bare-ice class
2	Debris-covered ice	Glacier ice covered by supraglacial debris

Labels are single-band UInt8 rasters, and value 0 is a valid background class. For binary glacier mapping, classes 1 and 2 can be merged into a glacier class. For debris-covered-ice identification, class 2 can be used as the target while retaining the distinction between bare ice and background.

3.2 Training and test split

Training samples originate from 40 source windows; test samples originate from the separately held-out windows 24 and 30. All three patch sizes retain this source split. No separate validation set is provided in the release directories. If validation data are needed, derive them from train using source-window or spatial groups, and keep test independent throughout model development.

4 Feature channels and numeric encoding

The four feature files contain 7, 3, 1 and 4 bands, respectively. Concatenating s2 → s1 → lst → dem produces a feature tensor of shape 15×N×N. The table below defines this concatenation order; band numbers within each file start at 1.

Ch.	File and band	Variable	Meaning and unit
1	s2 1	B2	Blue reflectance
2	s2 2	B3	Green reflectance
3	s2 3	B4	Red reflectance
4	s2 4	B8	Broad near-infrared reflectance
5	s2 5	B8A	Narrow near-infrared reflectance
6	s2 6	B11	Shortwave-infrared 1 reflectance
7	s2 7	B12	Shortwave-infrared 2 reflectance
8	s1 1	VV_dB	VV backscatter in dB
9	s1 2	VH_dB	VH backscatter in dB
10	s1 3	VV_minus_VH_dB	Difference between VV and VH in dB
11	lst 1	LST_C	Land surface temperature in °C
12	dem 1	elevation_m	Elevation in m
13	dem 2	slope_deg	Slope in degrees
14	dem 3	aspect_sin	Sine of aspect, dimensionless
15	dem 4	aspect_cos	Cosine of aspect, dimensionless

4.1 Data types and missing values

Folder	Data type	NoData code

Folder	Data type	NoData code
s2	Int16 or UInt16	-9999 or 32767
s1	Float32	-9999 or approximately -3.4028235×10^{38}
lst	Float32	-9999
dem	Float32	-9999
label	UInt8	No NoData value set

s2 stores scaled surface-reflectance integers; divide by 10,000 to recover reflectance, and do not automatically clip valid values to 0–1. s1 is already in dB, and its third channel is the difference between two dB values. lst is already in degrees Celsius; do not reapply the scale factor of the original Landsat product. Terrain variables are stored in the units listed above. The data have not been uniformly converted into standardized model-input tensors.

5 Data sources and patch construction

Optical features originate from Sentinel-2 Level-2A surface-reflectance products, radar features from Sentinel-1 GRD, temperature features from processing of Landsat 8/9 Collection 2 Level-2 data, and terrain variables from Copernicus DEM GLO-30. Reference masks are rasterized from the project's glacier vector labels using pixel-centre assignment, with debris-covered ice taking priority where label polygons overlap.

Features are organized on each window's Sentinel-2 reference grid in WGS 84 geographic coordinates, EPSG:4326, with an angular pixel step of approximately 0.0000898315284°. The "10 m" designation refers to the nominal grid scale; the actual ground dimensions of geographic pixels vary with latitude. Aligning the 20 m optical bands and coarser thermal-infrared and terrain data to this common grid does not add native spatial detail. Local alignment of continuous variables uses bilinear interpolation, while labels retain discrete class codes.

All three patch sizes are produced by extracting full-size crops from source windows. The interior stride equals the patch width. Where the remaining width or height at the right or bottom boundary is smaller than a full patch, a complete patch aligned to the window edge is added; edge patches may therefore overlap the preceding patch. Patches whose labels are entirely 0 are removed together with their four feature files, and the retained samples are renumbered consecutively. The sample pool contains glacier foreground but does not preserve the full distribution of pure-background patches.

6 Training and evaluation recommendations

Estimate normalization parameters, such as means, standard deviations or quantiles, from valid training pixels only, and apply them unchanged to validation and test data. Exclude missing values from these calculations. For models requiring dense inputs, missing values can be filled after normalization while retaining a validity mask. Apply geometric augmentation consistently to all features and labels, and use nearest-neighbour resampling for labels.

Different patch sizes share source windows and spatial coverage. Avoid mixing sizes and then randomly splitting patches across training and test sets. Pure-background removal and class imbalance affect false-positive assessment. Report performance separately for background, bare ice and debris-covered ice, and evaluate practical mapping on independent complete windows.

7 Access and citation

Obtain the three archives and this description from the AI-DCGs record in the data publication system. Creators, institutions, the formal release version, data identifiers, access conditions, licence and preferred citation are specified in that record. When reporting reuse, identify the patch size and training/test split used, and cite the dataset record and relevant source products.